

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform 4-space indentation used throughout all Python files.	Yes	PEP8 standard followed
2	Proper File Structure	Files organised into /models/, /api/, /dashboard/, /utils/, /tests/ folders.	Yes	Logical MVC-style layout
3	Meaningful Variable Names	Variables like nitrogen_level, crop_label, suitability_score reflect purpose.	Yes	Descriptive naming throughout
4	Function / Method Names	Functions named predict_crop(), check_suitability(), optimize_fertilizer() clearly.	Yes	Verb-noun convention used
5	Code Comments	Inline and block comments explain logic in all key functions and ML pipeline steps.	Yes	Docstrings added to all functions
6	Modular Design	Code split into crop_utils.py, soil_utils.py, ml_model.py, api_routes.py modules.	Yes	High reusability across modules
7	No Redundant Code	Duplicate logic removed; shared utilities centralised in utils/ package.	Yes	DRY principle applied
8	Error Handling	Try-except blocks handle invalid inputs, missing values, and model load failures.	Yes	Graceful error messages returned

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	crop_utils.py	Python / scikit-learn	Used in API, Dashboard & Testing modules	High
2	soil_utils.py	Python / pandas	Input preprocessing for recommendation & suitability checker	High
3	ml_model.py	Python / scikit-learn	Shared by recommendation & suitability endpoint	High
4	api_routes.py	Python / Flask	Exposes endpoints consumed by Dashboard & external clients	Medium
5	dashboard_components.py	Python / Streamlit	Reusable chart & form widgets across farmer & researcher views	Medium

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well-organised into clearly separated modules; follows MVC pattern.
Readability	5	Descriptive names, consistent formatting, and inline comments throughout.
Reusability	4	Core utility modules reused across API, UI, and tests; minor duplication in routes.
Documentation / Comments	4	All functions have docstrings; README and setup instructions are complete.
Overall Score	4.5	High-quality, production-ready codebase with strong modularity and readability.